



EVs: Towards a Greener Future

The sector needs to focus on infrastructure and innovation

By **Ajay Harinath Singh**

▶ **THE ELECTRIC VEHICLES (EVs)** segment is today playing a crucial role in decarbonising the transport sector, making it sustainable for the environment, economy, and ecology. Currently, India needs to focus on the five 'I's—Incentives (policy push), Innovation (technology), Investment (funding and financing), Infrastructure, and Integration (market creation). The government has taken several initiatives to create a robust EV ecosystem. The National Electric Mobility Mission Plan was launched in 2013. In 2019, the government approved the implementation of 'Faster Adoption and Manufacturing of Electric Vehicles in India Phase II (FAME India Phase II)' for the promotion of electric mobility. It aims to encourage faster adoption of EVs by offering an upfront incentive on purchase and by establishing a charging infrastructure. There is a plan to make available at least one charging station in a grid of 3 x 3 km. But soon, metros, cities and highways (where one charging station at a distance of 25 km is proposed) will need more than this.

INFRASTRUCTURE

There is an urgent need to have green sources for charging EVs to create a robust charging infrastructure. It is not practical under the current electricity regulations, which need a minimum of 1 MW load at the place of consumption to be eligible for Open Access power. For a robust charging infrastructure, it is prudent to lower the threshold of Open Access for sourcing power from green sources like solar. India has set a target of installing 500 GW of renewable-based energy capacity by the year 2030, a majority of which would come from solar and wind. The government needs to encourage power supply through Open Access.

INNOVATION IS THE KEY

The automobile industry needs to bring in advanced technology to make EVs efficient and cost-effective. The government needs to encourage advanced technologies and innovations. Benefits of incentives need to be extended to only those vehicles that are fitted with an advanced battery. Innovation in technology will play an important role in promoting a greener future but it demands huge investment. Big investments will help in

adopting a comprehensive approach, ranging from R&D to the production of green vehicles, which also includes strategic collaborations and partnerships.

INCENTIVES AND POLICY PUSH

India needs to leverage the government's ambitious 'Make in India' initiative and the industry should focus more on 'Make for the World' when it comes to EVs. The National Plan had set an ambitious target to achieve 6-7 million sales of hybrid and electric vehicles by the year 2020. According to Union transport ministry data, India had registered around 870,000 EVs in 2021 and it indicates a lot needs to be done to provide momentum to the green mobility sector.

Exemption of Goods and Services Tax (GST), currently charged at 5 per cent on vehicles and charging services, would provide a boost to the sector. The argument does not hold any value that under the new GST regime, the rates of GST on EVs have been kept in the lower bracket as against the 28 per cent GST rate with cess up to 22 per cent for conventional vehicles. For a greener future, the EV ecosystem needs to be free from GST.

THE WAY FORWARD

Financing the sector should be top priority. A report titled 'Banking on Electric Vehicles in India' by NITI Aayog also highlighted the importance of priority-sector recognition for retail lending in the electric mobility ecosystem. The inclusion of EVs in the Reserve Bank of India's priority-sector lending (PSL) can be a game changer.

Banks and non-banking financial companies in India have the potential to achieve an EV financing market size of ₹40,000 crore (\$5 billion) by 2025 and ₹3.7 lakh crore (\$50 billion) by 2030. However, retail finance for EVs needs an extra push. It would be wise for both the industry and financing agencies to begin with electric two-wheelers and three-wheelers. Gradually, the focus can be shifted to prioritise other categories under PSL. **BT**

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